Designed by Martin Wiese as part of the LLR Adult Cellulitis Teicoplanin Pathway

LRI Emergency Department

Cellulitis in adults

Version 57

Intended to aid severity classification & management

DO NOT **use in** **lymphedema** **or** **diabetic foot infections**

(use available alternative guidance), osteomyelitis, decubitus ulcer, chronic stasis ulcer or dermatitis

Disclaimer:
This is a clinical template; clinicians should always use judgment when managing individual patients

Re-approved by Antimicrobial Working Party 12Nov24 . Trust Ref: C44/2015
Review due Nov27

Patient details

Full name

DoB

Unit number

(use sticker if available)

① Features of instability?

☐ **YES** - at least one of the below

Acute physiology

- Pulse > 99/min (after antipyretics) ☐
- Respiratory rate > 20/min ☐
- Systolic BP < 100mmHg ☐
- Acutely altered mental state ☐

Comorbidity-related

- Chronic stasis ulcer ☐
- Uncontrolled diabetes ☐
- Peripheral vascular disease with critical ischaemia or ulcer ☐

☐ **NO** - none of the above

CBG

Vital signs

Temp

° C

SpO_2

% on air

Resp rate

per min

Heart rate

per min

BP

Flowchart

- Obtain vital signs & capillary blood glucose (CBG)
- Assess pain and address any analgesia needs
- Complete boxes 1 - 3

Are IV access and blood tests required (see box 4)?
- Y ☐ →
  - Obtain FBC, venous blood gas (VBG), U&E, LFT and coag screen
  - Record results in box 7
  - Complete Nervecentre Sepsis Assessment
- N ☐

- Offer antipyretic / antiemetics as needed
- Outline leading skin edge with indelible marker
- Document affected area on ED body map

Any sepsis red flags on NC assessment?
- Y ☐ → Red-flag sepsis likely → Class IV
- N ☐

Rapidly deteriorating cellulitis with purple discoloration and / or disproportional pain?
- Y ☐ → Necrotizing fasciitis is a concern → Class IV
- N ☐

Class IV

- Manage in the ER
- Complete sepsis six actions in Nervecentre assessment
- Inform DART on **#6826**
- If suspecting necrotizing fasciitis (see below), involve appropriate surgical specialty **NOW**
- In IVDU, consider anthrax

Features of instability (see box 1) or outpatient therapy exclusions (see box 5)?
- Y ☐ → **Class III** → Start antimicrobial as per box 6 on reverse
- N ☐

IV antimicrobials required (see box 3)?
- Y ☐ → **Class II** → Admit to EDU on Cellulitis pathway
- N ☐ → **Class I**

Enough support at home or able to self-care?
- Y ☐ →
  Complete discharge bundle
  - Prescribe TTO antimicrobials as per box 6 on reverse
  - Consider TTO analgesia
  - Hand out Cellulitis PIL
- N ☐ → Admit to EDU on Cellulitis pathway

Refer to Medicine (i.e. AMU/AFU/ACB) unless

- **Periorbital cellulitis** Ophthalmology
- **Other facial cellulitis** Maxillofacial surgeons

If suspecting necrotizing fasciitis

- **Scrotum** - Urology
- **Vulva** - Gynaecology
- **Trunk and perineum** (including axilla, groin and buttocks above infragluteal fold) – General Surgery
- **Lower Limb** (including buttocks below infragluteal fold) - Orthopaedics
- **Upper Limb**
  - Mon to Thu - Orthopaedics
  - Fri to Sun - Plastics
- **Head and Neck** – ENT

② Patient morbidly obese?

1. *Record weight and height below*
2. *Work out BMI using MD calc web calculator*

Weight

kg

Height

☐ cm ☐ in

BMI

☐ **YES**, as BMI is 40 or greater

☐ **NO**, as BMI is less than 40

③ IV antimicrobials required?

☐ **YES** – as at least one of the below

Systemic features of infection

- Temperature > 37.9° C ☐
- Vomiting ☐

Relevant comorbidities

- Morbid obesity (i.e. BMI ≥40; see box 2) ☐
- Treated diabetes ☐
- Glucose in ED > 11mmol/L ☐
- Liver cirrhosis ☐
- Evidence of peripheral vascular disease ☐
- Evidence of chronic venous insufficiency ☐

☐ **NO** – as none of the above

④ Blood tests needed?

☐ **Yes** – as at least one of the below

- Aged 65 years or older ☐
- Features of instability (see box 1) ☐
- IV antimicrobials required (see box 3) ☐

☐ **No** – as none of the above

⑤ Outpatient therapy exclusions?

☐ **Yes** – at least one of the below

- Cellulitis due to animal or human bite ☐
- Cellulitis known to be caused by MRSA ☐
- Facial or orbital involvement ☐
- Worsening while on outpatient IV antibiotics, or failure to improve after being on it for 48h ☐
- Rapidly progressive infection ☐
- Acute renal impairment (if U&E were needed) ☐
- Immunosuppression ☐
- Unrelated medical reason to admit ☐

☐ **No** – none of the above

Patient seen by

Print name	Signature	Role	Date	Time

⑥ Antimicrobial therapy recommendations *(mark the applicable regimen by ticking the relevant boxes)*

Important notes - read me first

- Seek microbiologist advice if cellulitis might be due to MRSA, or if patient is pregnant or breast-feeding
- If switching from flucloxacillin to teicoplanin there is no need to wait before first dose of teicoplanin
- Antimicrobials may enhance the effect of warfarin - increase INR monitoring during and after antimicrobial therapy

Severity class		Routine patients	Penicillin-allergic patients
I		☐ PO Flucloxacillin 1G QDS 1 week	☐ PO Doxycycline 200 mg OD for 1 week
II	Outpatient IV regimen (includes non-responders to class I therapy)	☐ **eGFR normal** • **Day 1** IV Teicoplanin BD ✱ • **Day 2-5** IV Teicoplanin OD ✱ • **Day 6-7** PO Flucloxacillin 1G QDS ☐ **eGFR 10 - 80mL/min** • **Day 1** IV Teicoplanin BD ✱ • **Day 2-4** IV Teicoplanin OD ✱ • **Day 5** - no antimicrobial - • **Day 6-7** PO Flucloxacillin 1G QDS ***eGFR < 10mL/min – patient not suitable***	☐ **eGFR normal** • **Day 1** IV Teicoplanin BD ✱ • **Day 2-5** IV Teicoplanin OD ✱ • **Day 6-7** PO Doxycycline 200mg OD ☐ **eGFR 10 - 80mL/min** • **Day 1** IV Teicoplanin BD ✱ • **Day 2-4** IV Teicoplanin OD ✱ • **Day 5** - no antimicrobial - • **Day 6-7** PO Doxycycline 200mg OD ***eGFR < 10mL/min – patient not suitable***
	Community hospital regimen	☐ **eGFR normal or > 9mL/min** • **Day 1-5** IV Flucloxacillin 2G QDS • **Day 6-7** PO Flucloxacillin 1G QDS ☐ **eGFR < 10mL/min** • **Day 1-5** IV Flucloxacillin 1G QDS • **Day 6-7** PO Flucloxacillin 1G QDS	☐ **eGFR normal** • **Day 1** IV Teicoplanin BD ✱ • **Day 2-5** IV Teicoplanin OD ✱ • **Day 6-7** PO Doxycycline 200mg OD ☐ **eGFR 10 - 80mL/min** • **Day 1** IV Teicoplanin BD ✱ • **Day 2-4** IV Teicoplanin OD ✱ • **Day 5** - no antimicrobial - • **Day 6-7** PO Doxycycline 200mg OD ***eGFR < 10mL/min – patient not suitable***
	✱ Teicoplanin dosing notes	☐ **under 71kg** give 400mg ☐ **71 - 100kg** give 600mg ☐ **101 - 130kg** give 800mg	☐ **131 - 170kg** give 1000mg ☐ **over 170kg** **discuss with microbiologist**
III & IV		☐ **eGFR normal** IV Flucloxacillin 2G QDS for 1 week ☐ **eGFR < 10mL/min** IV Flucloxacillin 1G QDS for 1 week	☐ Vancomycin for 1 week; dosing as per [Vancomycin Adult Prescription Chart] (print chart from 'ER – Other' [ED on-demand print menu])

⑦ Blood results

LFT

Albumin	
Bili	
AP	
ALT	

U&E

Na	
K	
Urea	
Crea	
eGFR	

Coagulation screen

INR	

FBC

WBC	
Hb	
Platelets	

⑧ Discharge vitals

SpO_2 on air	%
Resp rate	/min
Pulse rate	/min
BP	mm Hg
Temp	°C